1. Find the shortest string/s not in the language represented by the regular expression $a^*(ab)^*b^*$ (2)
   Answer: ba.
2. For the two regular expressions given below,
   a. find a string corresponding to r_2 but not to r_1 and (2.5)

      Answer: For example **ab** and **ba** are such strings.

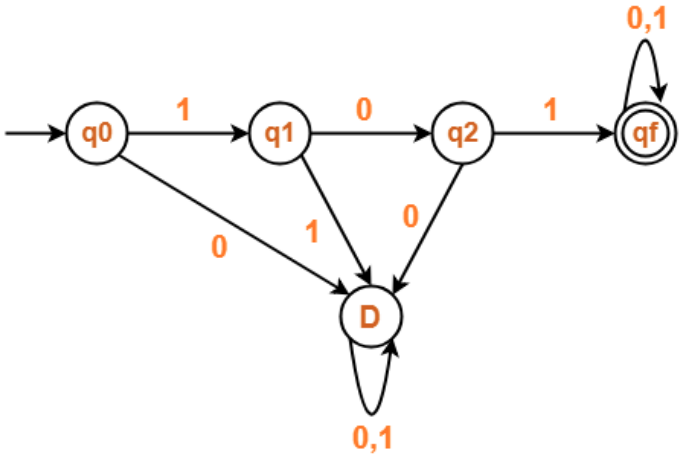
   b. find a string corresponding to both r_1 and r_2. (2.5)

      $r_1 = a^* + b^*$ $r_2 = ab^* + ba^* + b^*a + (a^*b)^*$
      Answer: a, b and the strings consisting of only b's (from $(a^*b)^*$).

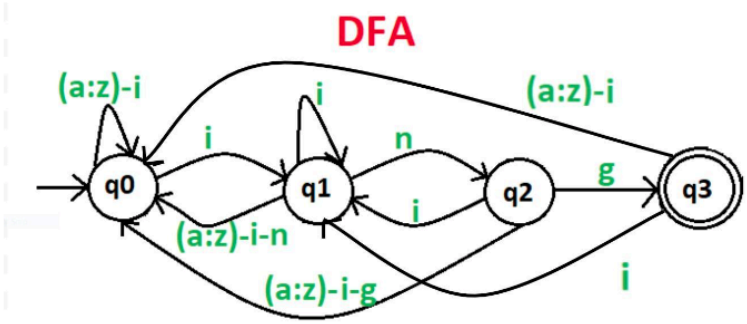
3. Find a regular expression corresponding to the language of strings of even lengths over the alphabet of { a, b } (5)

   Answer: $(aa + ab + ba + bb)^*$. Note that 0 is an even number. Hence the string ε is in this language.

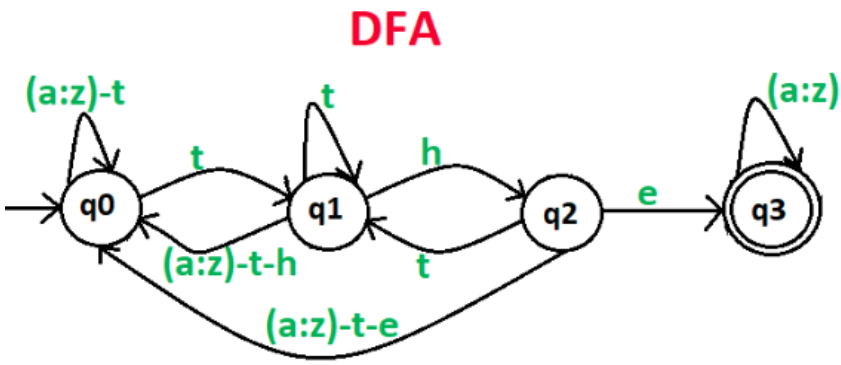
4. Draw a DFA for the language accepting 101(0 + 1)* (5)



5. Draw a DFA which accepts a string containing “ing” at the end of a string in a string of {a-z}, e.g., DFA will accept “anything” but not “anywhere”. (5)



6. Draw a DFA which accepts a string containing “the” anywhere in a string of {a-z}, e.g., DFA will accept “there” but not “those”. (5)



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7. Give context-free grammars that generate the following languages.

a. { w ∈ {0, 1} ∗ | w contains at least three 1s } (4)

Answer: $G = (V, \Sigma, R, S)$ with set of variables $V = \{S, X\}$, where S is the start variable; set of terminals $\Sigma = \{0, 1\}$; and rules

$$\begin{aligned} S &\rightarrow X1X1X1X \\ X &\rightarrow 0X \mid 1X \mid \varepsilon \end{aligned}$$

b. { w ∈ {0, 1} ∗ | w = w^R and |w| is even } (4)

Answer: $G = (V, \Sigma, R, S)$ with set of variables $V = \{S\}$, where S is the start variable; set of terminals $\Sigma = \{0, 1\}$; and rules

$$S \rightarrow 0S0 \mid 1S1 \mid \varepsilon$$